

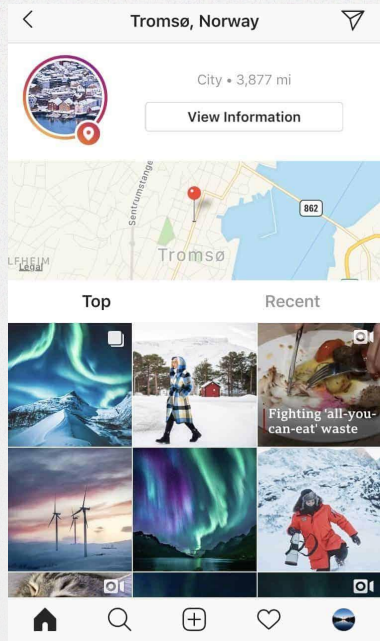


GeoMedia

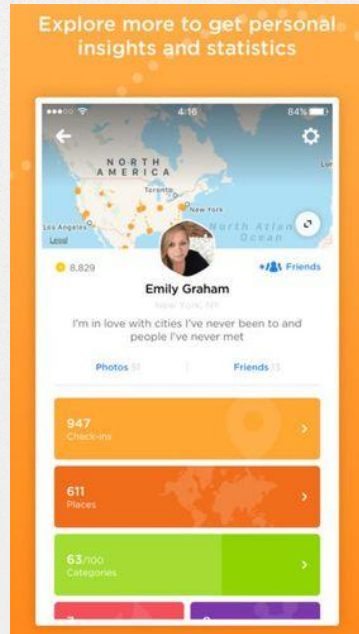
A Mobile Platform for Location-Aware
and Time-Sensitive Social Content

GoodIT 2025 | University of Padua, DD/MM/YYYY
Prof. Claudio Enrico Palazzi,

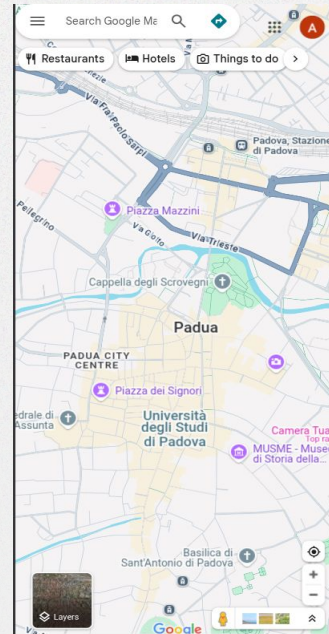
OSSN location based



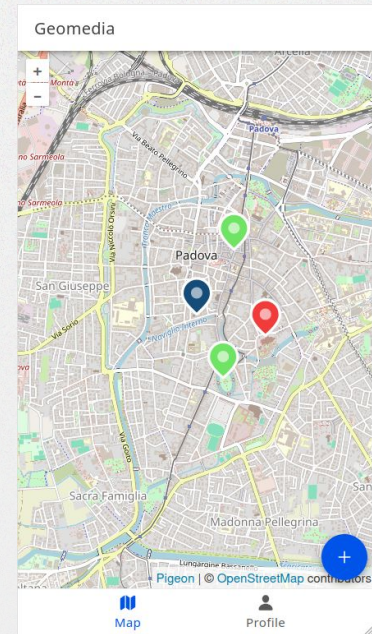
Instagram



Foursquare swarm

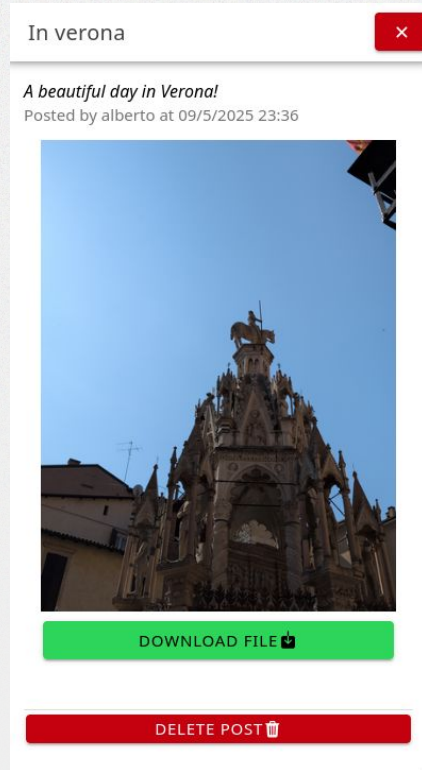
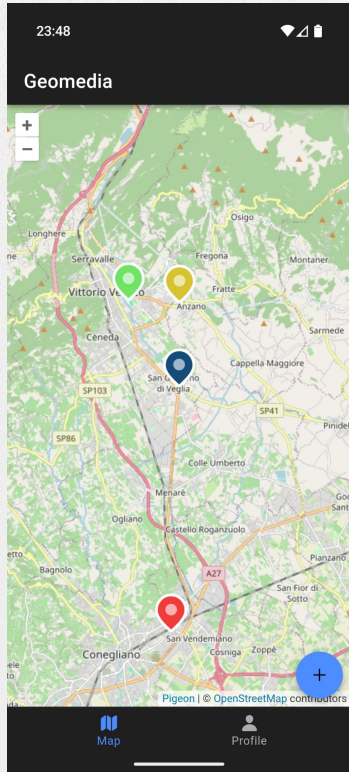


Google Maps



Geomedia

Panoramic of GeoMedia



Different kind of post markers:



The user current position



Post with just text



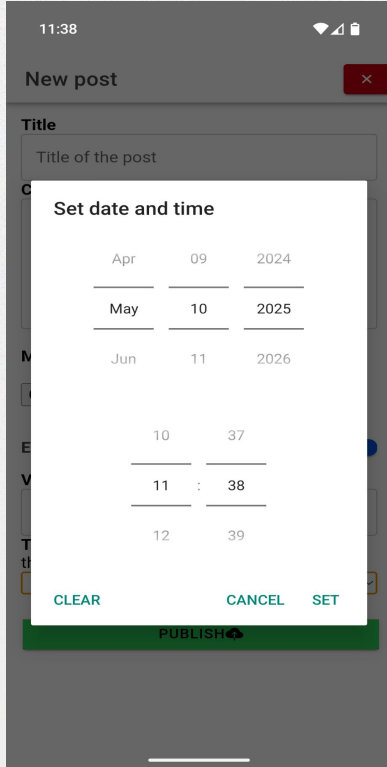
Post with media attachment (photos, videos, audio)



Exclusive content (by time or geographical area)

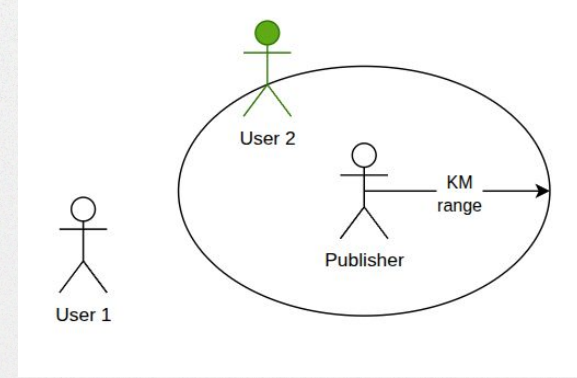
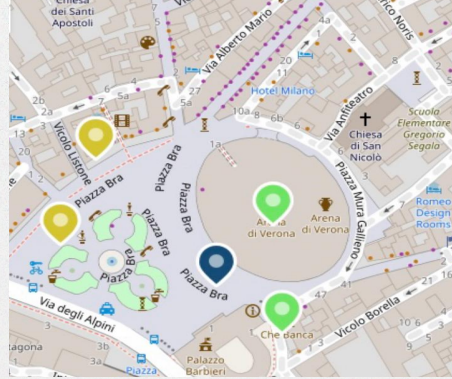
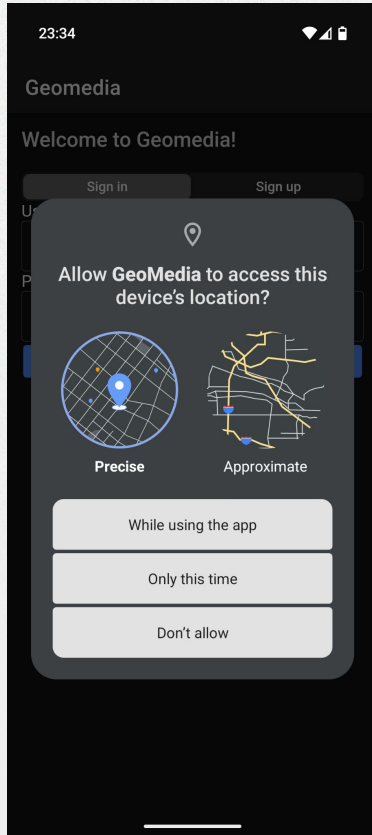
Hypermedia can be downloaded to the physical device file system.

Temporized post



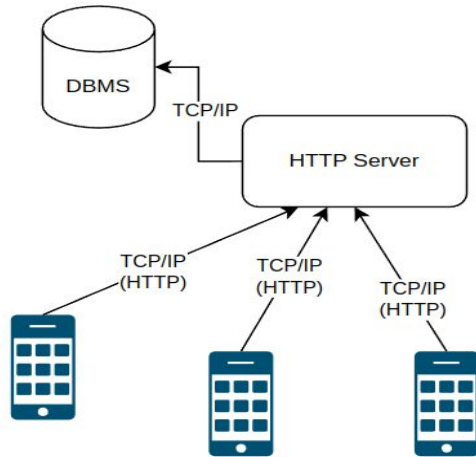
- Temporized posts will be reachable until the expiration date set by the publisher.
- The retrieving of the posts is made server side, which will compare the current datetime with the limit date of each temporized post

Geographical exclusivity

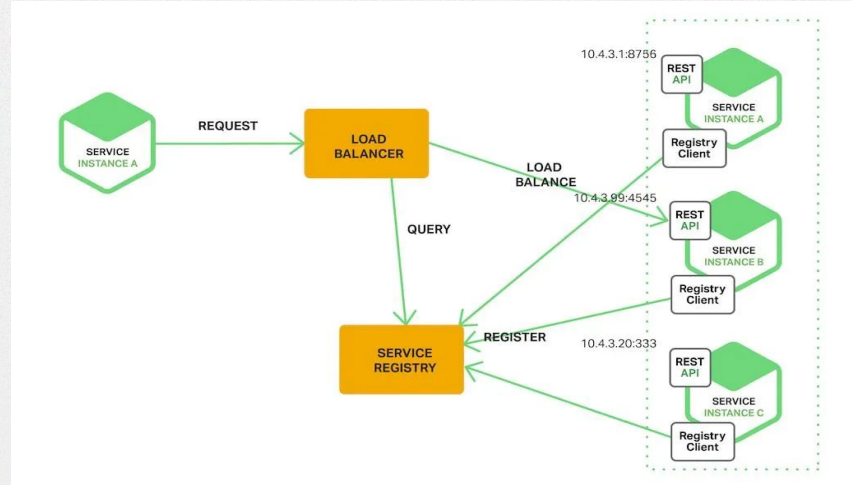


- Exclusive content based on location area is computed on real time within the actual end-user location
- Users when publishing can choose an exclusive area in km from their position in order to geographically limit the access

Architecture



Monolithic client-server
Adopted in GeoMedia

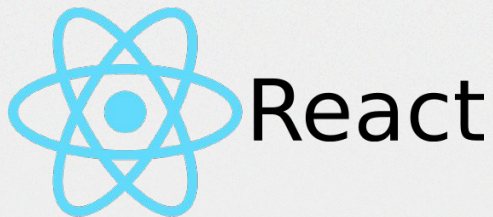


Example of microservice architecture, for better
scalability

Mobile application

Made with Ionic Framework and React.

Using Ionic Capacitor as interface to physical sensors such as GPS and file system.



Main purposes

- Provide UI and interact with users
- Get the current location of the end-users as coordinate
- Render posts in a map as markers
- Provide a profile section for users in order to see their post and settings (i.e. change server)
- Publish new content
- Fetch server with HTTP requests

Server side

Made with NodeJS

Using HTTP native library to
create a server

Using Tedious in order to connect
to a Microsoft SQL Server
database.



Main purposes

- Create an HTTP Server
- Retrieve and store data through the relative DataBase Management System
- Implement the Vincenty's formulae for distance computation of geographical exclusive posts

DataBase Management System

Made with Microsoft SQL Server



Main purposes

- Provide a relational database to store data in well defined tables
- Manage data with stored procedure and functions in order to preserve the single responsibility principle

Conclusion

In the end, GeoMedia offers

- A clean and simple interface to enrich a geographical map with content, allowing users to discover places near them
- The possibility to upload not just image or video, like other social networks usually do, but also audio, notes or any other file
- An Open Source online social network, privacy oriented without disclosing metadata about users to third actors

And a flexible framework to integrate other ideas and more features, for example:

- Time capsule function, storing data inside small sensors distributed among a city reachable with NFC or Bluetooth in order to be physically near
- Group games for events, such as “scavenger hunt” or “hide and seek”

Thank you!

Questions?